

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

November 2012

IT206 Object Oriented Programming using Java

Date: 8.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 Attempt Following. (Any Five)

[10]

1. What is java virtual machine?
2. Explain Bitwise shift operators with example.
3. What is interface? What is default modifier for an interface?
4. Explain two ways to create a Thread.
5. What is an Exception? Give four condition that generate an Exception.
6. What is stream? List out types of stream in java.

Q - 2 Answer the following.

- (a) Create a class called Circle. Provide default constructor, which initializes values for center coordinates and radius. Second constructor accepts only one argument for radius. And third constructor accepts three parameters. First two for center coordinates and the third parameter for radius. Also write print() method to print object of each type of constructor.

[05]

(b) **Attempt Following. (Any Two)**

[08]

1. What is inheritance? Explain different types of inheritance.
2. Explain basic features for java programming language.
3. What is package? Explain steps to create package with example.

Q - 3 Attempt following.

- (a) What is Applet? Draw and explain Life Cycle of Applet.

[08]

OR

- (a) What is Event? Explain event handling mechanism with suitable example.

[08]

(b) **Give the output of following code with explanation.**

[04]

1. class myclass

```
{  
    public static void main(String args[])  
    {  
        int a[] = { 4, 8, 16};  
        int i=1;  
        a[++i] = --i;  
        System.out.println(a[0] + a[1] + a[2]);  
    }  
}
```

```

    }
2. class A{
    int x=3;
    static int fun() { return x; }
}
class B {
    public static void main(String args[] )
    {
        System.out.println( A.fun( ));
    }
}

```

SECTION – II

Q - 4 Attempt following. (Any Two)

[10]

1. Write a java program using the concepts of File streams to implement concatenation of two files into third file. Use command line argument to read three file names. (File1.txt, File2.txt and File3.txt respectively)
2. Write an applet program to perform following.
Initiate applet window having one button labeled ON. When ON button is clicked label of that button is changed from ON to OFF. When the label of the button is OFF, and if button is clicked label of that button is changed from OFF to ON.
3. Declare an abstract class Vehicle with an abstract method named numWheels(). Provide subclasses Car and Truck that each implement this method. Create instances of these subclasses and demonstrate the use of this method.

Q - 5 Attempt following.

- (a) Draw the Use-case Diagram for withdraw traction process of ATM system.
- (b) Define Object and Class. Draw Class Diagram for "Inventory System".

[05]

[06]

OR

- (b) Define State Diagram. Draw State Diagram for "Phone Line System".

[06]

Q - 6

- (a) **Attempt any TWO.**

[08]

1. Define scenario. Write scenario for phone call.
2. Explain Object Oriented Methodology.
3. Give types of association with example.

- (b) **Attempt any THREE.**

[06]

1. How can encapsulation and polymorphism improve reusability?
2. Describe different ways to use data store.
3. Draw the notations of DFD.
4. Explain properties of aggregation with example.
